Image_Captioning

1. from [4]:
    1. Extract a 14x14 Feature Map from CNN.
    2. Image: $H \times W \times 3$ -CNN-> Features, v: $L \times D$, $L = W \times H$, this Feature vector becomes an input to become an RNN's first hidden state h_0. The RNN outputs a distribution over L possible locations, a_1. These are the locations of these grids in the feature vector $L \times D$. a_1 is multiplied with the feature map to get a weighted features $z_1 = \sum_{i=1}^{L} a_i v_i$ (v_i are the features inside each grid locations that come as the output of the feature map) which will become input together with y_1 (first word) or the start token (SOS) in the next RNN time step.
    3. h_1 which is the the next hidden step of the next time step in the RNN, gives us a_2 and d_1, a distribution over vocabulary that tells us what is the predicted word. a_2 is again multiplied with the feature map to give us a new weighted repr of the image feature map z_2. And d_1 could be the input y_2, or when we're training we're training we could use the second word of the caption to be input y_2 to the next step of the RNN, so z_2 and y_2 are the inputs to hidden state to the next step in the RNN. This generates a_3 and d_2. d_2 is the distribution over the vocabulary which tells us what is the word, a_3 tells us which part of the image to focus on. This is repeated again and again, and the caption is generated.
    4. In soft attention the a_1 the weight vector, is a softmax vector, and we use it as weights to multiply with each of grid locations in v. This gives us a soft attention or the feature map of the entire image.
    5. In hard attention we only choose a winning entry and only use that grid location as an input to the next time step. Hard since we're not using a soft distribution of the weights across all parts of the image.

References

1. Paper: An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition - Baoguang Shi et al
2. Convolutional Image Captioning, Jyoti Aneja, Aditya Deshpande, Alexander G. Schwig.
3. Karpathy et al, Deep visual-semantic alignments for generating image description, CVPR 2015.
4. Xu et al, Show attend and tell: Neural Image Caption generation with Visual Attention, ICML 2015.